

Principal Component Analysis (PCA)



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Synonyms

Karhunen–Loève transform (KLT)

Related Concepts

- [Dimension Reduction](#)
- [Eigenspace Methods](#)

Definition

Principal component analysis (PCA) is a standard tool in modern data analysis and is used by almost all scientific disciplines. The goal of PCA is to identify the most meaningful basis to reexpress a given data set. It is expected that this new basis will reveal hidden structure in the data set and filter out the noise. There are so many applications such as dimensionality reduction, data compression, feature extraction, and data visualization.

Background

Describe the background of the entry. The modern form of PCA was formalized by [1] who also introduced the term *principal component*. It is also known as the Karhunen–Loève transform.

Observations are often described by several dependent variables which are, in general, intercorrelated and include noise. PCA is used to extract the important information from such observations and to reduce the noise. To achieve this goal, PCA computes a set of new orthogonal variables called *principal components* which are obtained as linear combinations of the original variables. The values of these new variables for the observations are called *factor scores*. They can be interpreted as the projections of the observations onto the principal components. PCA can be also formulated as the linear projection that minimizes the average projection cost defined as the mean squared distance between the data points and their projections [2].

Several reformulations or extensions of PCA have been proposed. PCA can be expressed as the maximum likelihood solution of a probabilistic latent variable model [3]. This reformulation is known as *probabilistic PCA*. A nonlinear generalization of PCA, which is known as kernel PCA, is also proposed by using the approach of kernel learning [4]. Sparse principal component analysis finds sparse coefficients by introducing a constraint on the norm of the coefficients [5]. This sparseness makes the interpretation of the results of PCA easier.

Theory

Let $X = [\mathbf{x}_1, \dots, \mathbf{x}_N]$ be the data set to be analyzed by PCA, where each column is a single observation described by M variables. Then the sample mean vector $\bar{\mathbf{x}}$ and the sample covariance matrix Σ can be represented by

$$\bar{\mathbf{x}} = \frac{1}{N} \sum_{i=1}^N \mathbf{x}_i \quad (1)$$

$$\Sigma = \frac{1}{N} \sum_{i=1}^N (\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T = \frac{1}{N} \tilde{X} \tilde{X}^T \quad (2)$$

where the matrix $\tilde{X}$ is defined as $\tilde{X} = [\mathbf{x}_1 - \bar{\mathbf{x}}, \dots, \mathbf{x}_N - \bar{\mathbf{x}}]$.

To extract the important information from the observations, PCA computes factor scores as linear combinations of the original variables

$$y_{1i} = \mathbf{a}_1^T (\mathbf{x}_i - \bar{\mathbf{x}}), \quad (i = 1, \dots, N) \quad (3)$$

where $\mathbf{a}_1 = (a_{11}, \dots, a_{M1})^T$ is a set of weights of the linear combinations. The optimum weight vector $\mathbf{a}_1$ is obtained so that the sample variance of the new variable is maximized under the normalization constraint $\mathbf{a}_1^T \mathbf{a}_1 = 1$. Since the sample variance is represented by

$$V(y_{11}, \dots, y_{1N}) = \mathbf{a}_1^T \Sigma \mathbf{a}_1, \quad (4)$$

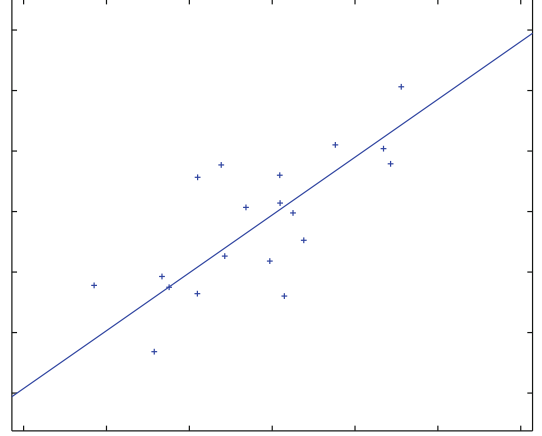
the optimization problem can be defined by using Lagrange multiplier λ_1 as the maximization of the Lagrange function

$$L(\mathbf{a}_1, \lambda_1) = \mathbf{a}_1^T \Sigma \mathbf{a}_1 - \lambda_1 (\mathbf{a}_1^T \mathbf{a}_1 - 1). \quad (5)$$

By setting the derivative of the Lagrange function with respect to the weight vector $\mathbf{a}_1$ equal to zero, the solution of this problem can be obtained as a unit eigenvector of the covariance matrix Σ corresponding to the largest eigenvalue λ_1 as

$$\Sigma \mathbf{a}_1 = \lambda_1 \mathbf{a}_1. \quad (6)$$

If the vector $\mathbf{a}_1$ is multiplied from the left, the maximum variance is given by



Principal Component Analysis (PCA), Fig. 1 An example of principal axis for two-dimensional samples

$$V(y_{11}, \dots, y_{1N}) = \mathbf{a}_1^T \Sigma \mathbf{a}_1 = \lambda_1. \quad (7)$$

Figure 1 shows an example of the first principal axis for the two-dimensional samples.

The second principal components

$$y_{2i} = \mathbf{a}_2^T (\mathbf{x}_i - \bar{\mathbf{x}}), \quad (i = 1, \dots, N) \quad (8)$$

can be defined as a new variable which maximizes the projected variance among all possible directions orthogonal to the first principal axis under the constraint on the normalization. The optimum coefficient vector $\mathbf{a}_2$ is also obtained as a unit eigenvector of the covariance matrix Σ corresponding to the second largest eigenvalue λ_2 . Similarly, additional principal components can be defined in an incremental fashion. For the general case of L -dimensional projection, the optimal linear projection of PCA

$$Y = A^T \tilde{X} \quad (9)$$

can be obtained by taking the L eigenvectors $A = [\mathbf{a}_1, \dots, \mathbf{a}_L]$ of the covariance matrix Σ corresponding to the L largest eigenvalue $\lambda_1, \dots, \lambda_L$. Then the eigenvector equation is given by

$$\Sigma A = A \Lambda, \quad (10)$$

where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_L)$.

PCA is closely related with the singular value decomposition (SVD) of the data matrix. The data matrix $\tilde{X}$ can be decomposed by SVD as

$$\tilde{X} = S\Delta V^T, \quad (11)$$

where S is the matrix of left singular vectors, V is the matrix of right singular vectors, and Δ is the diagonal matrix of singular values. By using this relation, the covariance matrix can be rewritten as

$$\Sigma = \frac{1}{N} \tilde{X} \tilde{X}^T = \frac{1}{N} S \Delta V^T V \Delta S^T = \frac{1}{N} S \Delta^2 S^T. \quad (12)$$

By multiplying S from the right, the eigenvector equation is obtained as

$$\Sigma S = S \frac{1}{N} \Delta^2. \quad (13)$$

This shows that the weight vectors A are equal to S and the diagonal matrix of the eigenvalues Λ is equal to $\frac{1}{N} \Delta^2$ if the number of principal components L is equal to the rank of the data matrix $\tilde{X}$. From these relations, the matrix of factor scores Y can be represented as

$$Y = A^T \tilde{X} = S^T S \Delta V^T = \Delta V^T. \quad (14)$$

This shows that the principal components are also obtained from the SVD of the data matrix $\tilde{X}$. By using this relation, the data matrix $\tilde{X}$ can be represented as the product of the score matrix by the weight vectors A :

$$\tilde{X} = S \Delta V^T = A Y = A A^T \tilde{X}. \quad (15)$$

This means that the original data matrix $\tilde{X}$ can be reconstructed from the factor scores.

If the number of principal components L is less than the rank of $\tilde{X}$, this reconstruction gives

an approximation of the original data matrix. The mean squared errors between the original observations and the approximations are then given by

$$\begin{aligned} \varepsilon^2(L) &= \frac{1}{N} \sum_{i=1}^N ||(\mathbf{x}_i - \bar{\mathbf{x}}) - A A^T (\mathbf{x}_i - \bar{\mathbf{x}})||^2 \\ &= \sum_{i=1}^{\text{rank}(\tilde{X})} \lambda_i - \sum_{i=1}^L \lambda_i \end{aligned} \quad (16)$$

which is simply the sum of the $\text{rank}(\tilde{X}) - L$ smallest eigenvalues. This means that PCA is minimizing this approximation errors by selecting the principal subspace spanned by the eigenvectors corresponding to the L largest eigenvalues.

Application

There are many applications of PCA in pattern recognition or computer vision. One of the famous applications of PCA is for face recognition [6]. Face images are decomposed into a set of characteristic feature images called “eigenfaces,” which are the eigenvectors computed by PCA to the training set of face images. Recognition is performed by projecting a new face image into the subspace spanned by the eigenfaces and then classifying the face by comparing the distances of factor scores. Figure 2 shows examples of the eigenfaces computed from 200 face images.

PCA is also used to construct 3-D object models from their appearances [7]. To obtain a low-dimensional subspace of object appearance parameterized by pose and illumination, PCA is applied to a large set of images obtained by varying pose and illumination. Then a new image is projected to the constructed subspace, and the



Principal Component Analysis (PCA), Fig. 2 Examples of eigenfaces computed from 200 face images

recognition is performed on the subspace. The object's pose in the image is also determined from the position of the projection on the subspace.

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